August 5, 2026

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer remote contract opportunity with Crossing Hurdles. With a Master of Science in Computer Science from the University of Calgary and applied experience in machine learning research, PyTorch experimentation, model evaluation, and technical writing, I am excited by the chance to help design and assess high-quality machine learning engineering benchmarks.

My graduate research focused on developing scalable machine learning and data-management solutions for large geospatial and Earth observation datasets. I designed, trained, and evaluated deep neural networks in Python and PyTorch, built reproducible pipelines for data cleaning and model validation, and documented technical decisions carefully enough for publication, thesis work, and research review.

One of my primary research projects examined the reliability and generalizability of deep-learning models for digital elevation model super-resolution. I implemented convolutional neural-network training workflows and evaluated model outputs using statistical and terrain-specific metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. Through systematic experimentation, I identified how random data splitting could produce overly optimistic results and used region-based validation to better assess real-world performance. This background maps directly to work that requires careful judgment about model training, debugging, optimization, evaluation, and experimental design.

My second major research area involved developing a hierarchical framework for managing and retrieving large volumes of spatiotemporal satellite data. This work required integrating data from multiple sources, designing multiresolution processing methods, assessing storage and retrieval performance, and communicating results in a first-author peer-reviewed publication. It strengthened my ability to translate complex technical workflows into structured, defensible outputs.

I also bring experience evaluating and explaining technical work. As a teaching assistant at the University of Calgary and a Deep Learning Teaching Assistant with Neuromatch Academy, I guided students through PyTorch tutorials, CNNs, transformers, reinforcement learning, research projects, debugging, and final presentations. These roles developed the written communication, attention to detail, and independent judgment needed for asynchronous benchmark and assessment work.

The short-term, remote, research-driven nature of this opportunity is a strong fit for how I work. I am comfortable operating independently, reviewing model behavior and technical tradeoffs, writing precise evaluations, and delivering high-quality outputs on defined timelines.

Thank you for considering my application. I would welcome the opportunity to contribute my machine learning research background, evaluation mindset, and technical writing skills to Crossing Hurdles' remote machine learning engineering work.

Sincerely,
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